A Diffusion Model-Based Suppression Method for Complex and Unseen SAR RFI

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***Abstract*—In the increasingly complex electromagnetic environment, radio frequency interference (RFI) poses a significant challenge to synthetic aperture radar (SAR) systems, degrading image quality and corrupting valuable data. Existing deep learning-based suppression methods, which typically learn a direct mapping from contaminated data to clean data, are limited to the interference types encountered during training, leading to poor generalization to unknown RFI patterns and novel scenes. To address these challenges, this article proposes a diffusion model-based method through reformulating RFI suppression as a conditional generative task. Conditioned on the uncorrupted information, the proposed model executes generative restoration of contaminated data by learning the distribution of clean SAR signals in the time–frequency domain. This generative paradigm enables the model to be robust to complex and unseen RFI patterns, as it aims to learn the clean data distribution rather than to recognize specific RFI types. To further enhance generalization across diverse RFI scenarios, the model is trained using a library of physically simulated RFI masks. Besides, we also introduce a composite loss function that incorporates the low-rank property of SAR data to enforce structural consistency. Experiments tested on Sentinel-1 data confirm the method's superiority in complex and unknown RFI suppression, especially for the geographical regions that were completely unseen during training.**

***Index Terms*—Denoising diffusion probabilistic model (DDPM), generative model, radio frequency interference (RFI), Sentinel-1, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active microwave remote sensing instrument that can acquire high-resolution imagery of the Earth's surface under all-weather and all-day conditions, in contrast to optical cameras [1], [2], [3], [4], [5]. SAR systems can be mounted on space vehicles, offering advantages such as global coverage, periodic revisits, and multidimensional observation capabilities. These features have led to widespread use in diverse fields, including natural disaster detection, ocean monitoring [6], [7], and military reconnaissance [8]. However, as the radio frequency (RF) spectrum has become increasingly crowded with the growth of wireless communication technologies, RF interference (RFI) from other radiating sources [9] has become a significant source of contamination for SAR imaging. In focused spaceborne SAR images, RFI can manifest as large shadowed regions, bright artifacts, or ghost targets, which severely degrade subsequent applications like target recognition, image interpretation, and change detection.

Numerous methods have been developed so far to mitigate RFI in SAR data, which can be broadly classified into three categories: parametric, nonparametric, and semi-parametric. Parametric methods construct precise mathematical models of the interference, and estimate its parameters using techniques such as least-squares [10], iterative adaptive approach (IAA) [11], [12], RELAX algorithm [13], and fractional Fourier transform (FRFT) [14]. For instance, Yang et al. [15] analyzed the linear frequency modulated (LFM) properties of RFI in single look complex (SLC) SAR images, proposing 2-D SPECAN for its removal. However, the time-varying and nonstationary nature of RFI in real-world electromagnetic environments can lead to model mismatch and a significant increase in computational burden. Nonparametric methods exploit the distinct features that differentiate RFI from desired SAR signals, including notch filtering [16], [17], eigenvalue subspace projection (ESP) [18], [19], and sub-band spectral cancellation (SSC) [20]. Daković et al. [21] designed a time–frequency notch filter (TFN) to remove nonstationary interference by masking concentrated energy regions in the time–frequency domain. Recently, Li and Hu [22] proposed a 2-D dynamic threshold ESP (2DDT-ESP) method to mitigate mutual RFI (MRFI) between SAR satellites. Mao et al. [23] utilized a frequency-domain notch filter (FNF) within a hybrid framework to mitigate extremely strong RFI while protecting scene details. While effective, nonparametric methods inevitably cause a loss of the useful signal, thereby degrading the quality of the final focused SAR image. Semi-parametric methods separate the RFI components by solving an optimization problem. The most widely used model is the low-rank and sparsity model, which is commonly combined with robust principal component analysis (RPCA) [24], [25], [26], [27], [28], [29]. Nevertheless, these methods typically demand extensive iterative computation and are sensitive to hyperparameter tuning. Furthermore, their performance

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degrades in complex electromagnetic environments with diverse RFI types or low signal-to-interference-plus-noise ratios (SINRs), where the underlying low-rank and sparsity assumptions are weakened [30].

In recent years, deep learning has emerged as a powerful paradigm for SAR RFI suppression, leveraging its powerful capabilities in nonlinear feature extraction [31]. Inspired by image denoising in computer vision, Wei et al. [32] introduced a combined attention reconstruction network (CARNet) with a feature attention block (FAB) to improve image details during reconstruction. Similarly, Li et al. [33] developed a multidimensional calibration and suppression network (MCSNet) to maintain scattering characteristics by extracting deep image features. One key drawback of these networks is that they operate directly on real-valued magnitude SAR images, discarding crucial phase information, and their performance degrades significantly under low SINR conditions. In contrast, numerous studies operate in the time–frequency domain (TFD) [34], which provides a richer feature representation for separating RFI and SAR signals. For instance, the interference mitigation network (IMN) [35] used a ResNet-based architecture to transform a contaminated time–frequency spectrum into a clean one. Wang et al. [36] proposed a dual-path residual network with the attention mechanism for Comb Spectrum Modulation Jamming (CSMJ) suppression. Shen et al. [37] proposed a prior-induced interference suppression network (PISNet) based on the low-rank characteristics of TFD. More recently, Transformer-based architectures have been used to segment RFI in the spectrogram and then inpaint the corrupted regions [38]. Xu et al. [39] first employed generative adversarial networks (GANs) to suppress wideband interference. However, this approach is limited by training instability and restriction to a single type of interference. Although these networks have a relatively straightforward architecture, they are trained to learn a deterministic mapping from the contaminated to the clean data, which presents two critical limitations. First, they exhibit poor generalization to unknown interference. Since the models are trained on datasets containing specific simulated RFI types, their performance degrades significantly when confronted with complex, time-varying mixed interference or novel interference patterns encountered in real-world electromagnetic environments. Second, they exhibit high data dependency. Their performance deteriorates severely when applied to new geographical areas whose statistical characteristics differ from the training dataset.

To address these challenges, this article reformulates SAR RFI suppression as a conditional generation task and introduces the denoising diffusion probabilistic model (DDPM), also known as diffusion models [40]. Unlike conventional deep models that learn a single mapping, the diffusion model aims to learn the underlying data distribution of clean SAR signals. During inference, conditioned on uncontaminated data regions, the model performs generative restoration within the corrupted regions. This distribution-based restoration principle overcomes the limitations of previous methods effectively. To generate signals consistent with the learned distribution, the model is not restricted to the specific RFI types seen during training, thus exhibiting inherent robustness against complex and unseen interference. More importantly, by capturing the generative mechanisms of SAR signals and leveraging robust spatial context reasoning, the model can generalize to new geographical areas as well, thereby overcoming the data dependency of previous methods.

To further improve the model's generalization performance and restoration quality, we also introduce a spatial prior as a training strategy, which consists of a diverse, physically simulated library of RFI masks. Training with these masks can simultaneously force the model to learn restoration under various complex RFI patterns, thereby enhancing its generalization ability against diverse real-world interference. In addition, we design a novel composite loss function that incorporates a low-rank property to enhance structural consistency in generated signals.

Overall, the main contributions of this article are threefold.

1) We propose a novel SAR RFI suppression framework based on a conditional diffusion model, which reformulates the task as a conditional generation problem. By learning the underlying data distribution of clean SAR signals, this method addresses the limitations of existing methods in suppressing complex and unknown RFI types.
2) We design a novel training strategy that integrates spatial prior to enhance the model's generalization. A composite loss function that incorporates the low-rank property of TFD is correspondingly constructed to enforce the structural consistency as well.
3) Extensive experiments are performed on Sentinel-1 dual-polarization SAR data, which demonstrate the superiority and generalization of the proposed method. Meanwhile, the experimental results also show that our method can maintain better suppression performance on the data obtained from new geographical regions.

In the following sections, Section II introduces the signal models and diffusion model. Section III details the proposed RFI suppression method. Section IV presents experimental results and analysis, and Section V concludes the article.

II. Theoretical Background and Preliminaries

In this section, the signal models of SAR echo and RFI are presented, followed by an analysis of their characteristics in the TFD. Subsequently, the fundamental principles of DDPM are introduced, which serve as the core of our proposed method.

A. Signal Model and Characteristics Analysis

SAR systems are susceptible to RFI from various sources operating in the same frequency band, such as ground-based radars and television transmitters. The total received signal $Y(\tau,\eta)$ can be expressed as

$$Y(\tau,\eta) = S_r(\tau,\eta) + I(\tau,\eta) + N(\tau,\eta) \tag{1}$$

where $S_r(\tau,\eta)$ is the clean echo, $I(\tau,\eta)$ is the RFI signal, and $N(\tau,\eta)$ is the background noise. $S_r(\tau,\eta)$ can be expressed as

$$S_r(\tau,\eta) = \iint_{(x,y)\in\Omega} \sigma(x,y)\, s_0(\tau,\eta)\, dxdy \tag{2}$$

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where τ and η represent fast and slow time, respectively, Ω denotes the observed scene, and $\sigma(x, y)$ is the backscattering coefficient of the scene at position (x, y). $s_0(\tau, \eta)$ is the baseband echo which can be formulated as

$$s_0(\tau,\eta) = \omega_r(\tau - 2R(\eta)/c)\,\omega_a(\eta - \eta_c) \times \exp(-j4\pi f_0 R(\eta)/c)\exp\left(j\pi K_r(\tau - 2R(\eta)/c)^2\right) \tag{3}$$

where $\omega_r(\cdot)$ and $\omega_a(\cdot)$ are the range and azimuth window functions, $R(\eta)$ is the slant range model used in classical SAR imaging theory, η_c is the beam center crossing time, f_0 is the carrier frequency, K_r is the range chirp rate, and c is the speed of light.

RFI is generally classified by its bandwidth into narrowband interference (NBI) and wideband interference (WBI) [38], [41], [42], [43], [44]. An NBI signal, which is a composite of multiple sinusoids, can be modeled as

$$I_{nbi}(\tau,\eta) = \sum_{n=1}^{N} A_n(\eta)\exp(j(2\pi f_n\tau + \phi_n)) \tag{4}$$

where $A_n(\eta)$, f_n, and ϕ_n are the amplitude, carrier frequency, and the initial phase of the nth NBI, respectively. WBI is often categorized as linear frequency-modulated (LFM) and sinusoidal-modulated (SM). The LFM WBI is expressed as

$$I_{lfm}(\tau,\eta) = \sum_{n=1}^{N} A_n(\eta)\exp\left(j\left(2\pi f_n\tau + \pi K_n\tau^2 + \phi_n\right)\right) \tag{5}$$

and the SM WBI is modeled as

$$I_{sm}(\tau,\eta) = \sum_{n=1}^{N} A_n(\eta)\exp(j2\pi f_n\tau) \times \exp(j\beta_n\sin(2\pi f_{mn}\tau + \phi_n)) \tag{6}$$

where K_n, β_n, and f_{mn} are the chirp rate, modulation factor, and modulation frequency, respectively.

Another important RFI category is pulse RFI (PRFI), which exhibits intermittent transmission characteristics [43], [45], [46], [47]. This characteristic is modeled by modulating the continuous RFI with an additional rectangular window function, $\mathrm{rect}(\tau/T_d)\mathrm{rect}((\tau - \tau_I(\eta))/T_I)$, where T_I and τ_I are the pulsewidth and central time of PRFI, and T_d is the duration of the SAR system reception.

In deep learning-based suppression methods, raw SAR echo is typically transformed into the TFD via the short-time Fourier transform (STFT), as SAR echoes and RFI exhibit distinct and stable feature representations in this domain. For a discrete signal $x(n)$, its STFT matrix can be expressed as

$$STFT(m,k) = \sum_{n=0}^{N-1} x(n)\,h(m-n)\exp(-j2\pi kn/N) \tag{7}$$

where $h(m)$ is the window function, N is the length, and k is the discrete frequency index. The signal $x(n)$ can be reconstructed by inverse STFT (ISTFT).

Based on these models, the distribution of RFI in TFD is determined by its parameters. NBI manifests as horizontal lines, indicating a constant frequency over the time axis; the parameter f_n determines the vertical position of these lines in the matrix. LFM WBI appears as inclined lines representing a linearly varying frequency, with the chirp rate K_n determining the slope. SM WBI presents as a sinusoidal curve, with its shape governed by the modulation parameters β_n and f_{mn}. The properties of the STFT itself, such as the length of the window function $h(m)$ and the overlap between adjacent windows, also affect the resolution and the final representation of the RFI.

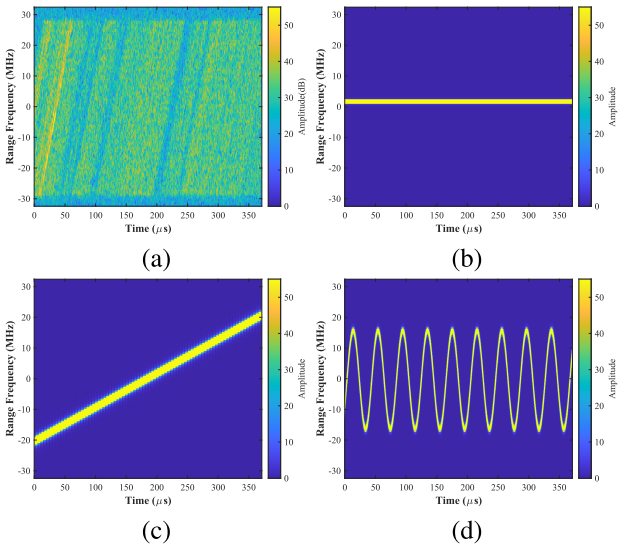

Fig. 1. TFD representations of an uncontaminated SAR echo and three typical types of RFI. (a) Uncontaminated SAR echo. (b) NBI. (c) LFM WBI. (d) SM WBI.

Fig. 1 illustrates the TFD representation of an uncontaminated Sentinel-1 SAR echo alongside three typical RFI signals. As derived from the models in (2) and (7), the clean SAR echo in Fig. 1(a) is a superposition of many LFM signals resulting from bidirectional propagation and scattering off ground targets. Consequently, the clean echo manifests as a composite of slanted linear structures whose amplitude distribution reflects the scattering coefficients of the terrain illuminated by the radar beam. In contrast, the high-energy, one-way RFI signals shown in Fig. 1(b)–(d) exhibit simpler and more distinct geometric features. This significant difference in energy distribution and geometric structure between RFI and SAR echoes in the TFD provides the fundamental basis for RFI detection and localization.

B. Denoising Diffusion Probabilistic Model

Generative models are a powerful class of models that can learn the underlying distribution of a given dataset and generate new samples statistically indistinguishable from the original data by sampling from the learned distribution, such as variational autoencoder (VAE) [48] and GAN [49]. Recently, DDPM [40], also known as the diffusion model, has gained prominence for its powerful distribution modeling and generation capabilities. Diffusion models have already been applied in various fields, including SAR target generation [50], object detection [51], and multimodal fusion [52].

The training of the diffusion model involves two phases: forward diffusion and reverse denoising. In the forward stage, for a given clean image x_0, a sequence of noisy images

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$x_1, x_2, \ldots, x_T$ is generated through T steps of Gaussian noise addition, which can be modeled by a Markov chain

$$q(x_t|x_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}x_{t-1}, \beta_t \boldsymbol{I}\right) \tag{8}$$

where $q(x_t|x_{t-1})$ is the transition probability distribution after adding Gaussian noise at each step, $\beta_t \in (0, 1)$ represents the variance parameter that increases linearly with timestep t, and $\boldsymbol{I}$ is the identity matrix. In the forward process, the noisy image x_t can be directly derived from the original image x_0 as

$$\begin{aligned} q(x_t|x_0) &= \mathcal{N}\left(\sqrt{\bar{\alpha}_t}x_0, 1-\bar{\alpha}_t \boldsymbol{I}\right) \\ x_t &= \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon \end{aligned} \tag{9}$$

where $\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s)$, $\epsilon \sim \mathcal{N}(0, \boldsymbol{I})$. When $T \to \infty$, x_T converges to a standard Gaussian distribution.

The reverse denoising stage aims to obtain the transition probability distribution $q(x_{t-1}|x_t)$. In practice, a deep model is employed to learn the distribution $p_\theta(x_{t-1}|x_t)$ and restore the image $\hat{x}_0$ from the noise step by step as $q(x_{t-1}|x_t)$ is difficult to obtain, $p_\theta(x_{t-1}|x_t)$ can be expressed as

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}\left(\mu_\theta(x_t, t), \Sigma_\theta(x_t, t)\right) \tag{10}$$

where θ denotes the parameters of the model. Ho et al. [40] propose using a network to estimate the noise $\epsilon_\theta(x_t, t)$ that needs to be removed from x_t to resotre x_0, where the mean can be expressed as

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\bar{\alpha}_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t)\right) \tag{11}$$

and the loss function for training the model is defined as

$$L_{\text{simple}} = \mathbb{E}_{x_0,t,\epsilon}\left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2\right]. \tag{12}$$

Essentially, this loss function optimizes the network parameters θ by minimizing the distance between the predicted noise $\epsilon_\theta(x_t, t)$ and the true noise ϵ.

III. Proposed Method

A. Overview

High-performance SAR RFI suppression method aims to precisely remove interference and restore reliable and coherent useful signals within the contaminated regions. In TFD, this problem can be reframed as a conditional generation task: restoring corrupted regions conditioned on the surrounding uncontaminated signals. However, this task is different from conventional optical image inpainting. SAR data in TFD exhibits unique structural characteristics due to the working principles of SAR systems.

Based on these considerations, we propose an SAR RFI suppression framework based on a diffusion model. By learning the underlying distribution of clean SAR signals, the model generates restorations that are consistent with this learned distribution, thereby granting it generalization against unknown and complex interference types. The overall workflow is illustrated in Fig. 2. The method begins by transforming the raw SAR echo into TFD using STFT. Subsequently, a segmentation network is employed to precisely localize the RFI-contaminated regions. Since RFI typically exhibits simple geometric structures and high-energy features in the TFD, we find that a standard U-Net architecture is sufficient for this task. Conditioned on the uncorrupted regions, the diffusion model performs a generative restoration within the contaminated areas. The processed TFD data are then transformed back to the time domain using ISTFT for final imaging.

B. RFI Localization via Dual-Polarization Fusion

To accurately condition the diffusion model, RFI must first be precisely localized. For this goal, we leverage a key physical property of dual-polarization SAR systems.

Modern dual-polarization systems like Sentinel-1 employ a single-transmit, dual-receive architecture where two orthogonal channels receive echoes simultaneously. This results in a crucial characteristic: while the useful echo energy differs between channels due to polarimetric scattering, the RFI signature, originating from an external source, exhibits a high degree of consistency in its geometric shape, start time, and spatial position within TFD. This cross-channel geometric alignment, clearly demonstrated with real Sentinel-1 data in Fig. 3, provides a theoretical basis for our approach.

To exploit this property, we design a dual-polarization fusion U-Net (DPF-UNet), shown in Fig. 4. The network accepts a three-channel tensor input: VH polarization time–frequency representation (TFR) amplitude ($|\mathbf{TF}_{VH}|$), VV polarization TFR amplitude ($|\mathbf{TF}_{VV}|$), and their conjugate product ($|\mathbf{TF}_{VH} \odot \mathbf{TF}^*_{VV}|$), where $\odot$ denotes Hadamard product. In addition, the third channel, created from the cross-channel conjugate product [53], uses the interchannel correlation to strengthen the structural features of RFI while suppressing uncorrelated background echo, making this channel a strong feature for the network to identify RFI. Architecturally, DPF-UNet uses a U-Net structure and efficient ConvNeXt blocks [54] as its feature extraction backbone. The network outputs a binary segmentation mask $\mathbf{M}_{RFI} \in \mathbb{R}^{H\times W}$, where 1 represents the presence of RFI, and 0 represents the absence.

C. Restoration by Conditional Diffusion Model

The mask $\mathbf{M}_{RFI}$ produced by DPF-UNet partitions TFD into uncontaminated regions and corrupted regions that need restoration. This allows us to define the RFI suppression problem as a conditional generation task: generating the signal within the corrupted regions conditioned on the surrounding uncontaminated data. To address this task, we employ DDPM as the core generative model. Sections III-C1–III-C3 and III-D detail the key components of our conditional diffusion framework, including its formulation, training strategy, composite loss function, and inference process.

1) Conditional Restoration Formulation: The framework of the conditional diffusion model training is illustrated in Fig. 5. Given a clean TFR sample $\mathbf{X}_0 \in \mathbb{R}^{2\times H\times W}$, the conditional data $\mathbf{X}_{cond}$ can be expressed as

$$\mathbf{X}_{cond} = \mathbf{X}_0 \odot (1 - \mathbf{M}_{RFI}) \tag{13}$$

where the RFI regions are set to zeros. The forward diffusion process progressively adds Gaussian noise $\epsilon \sim \mathcal{N}(0, \boldsymbol{I})$ to the

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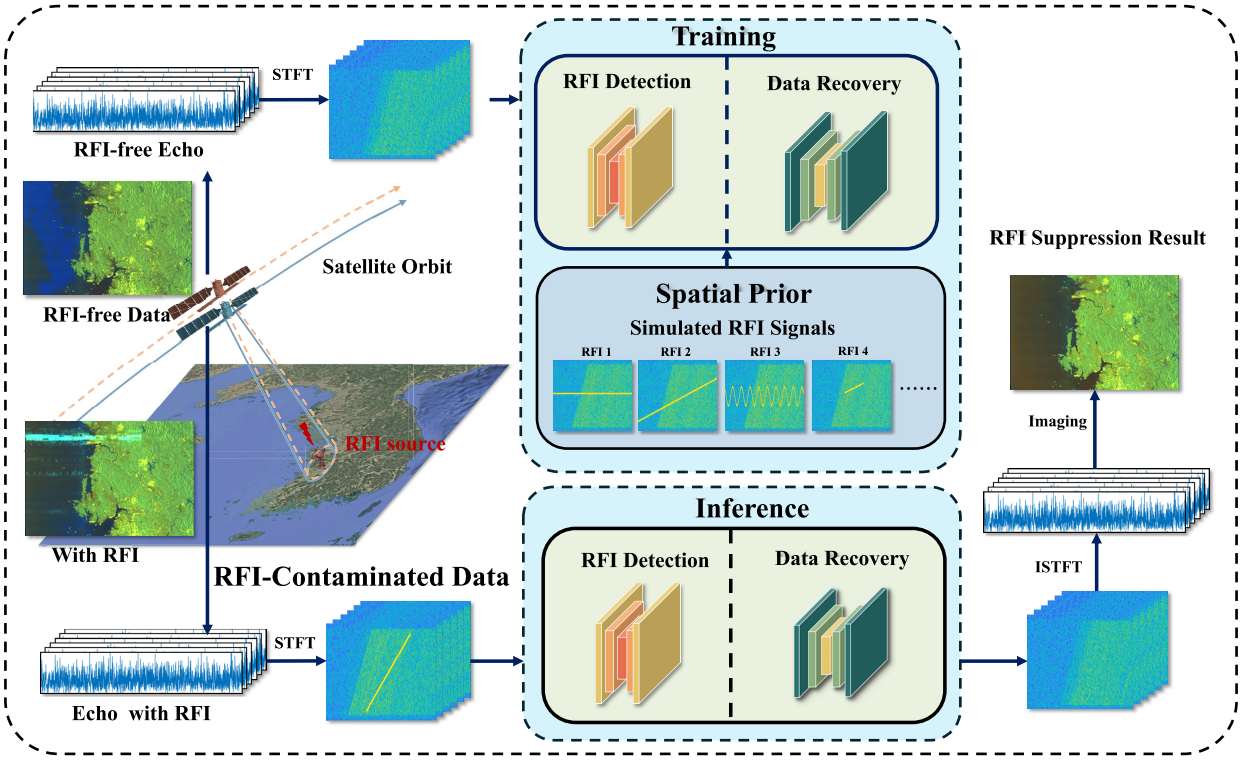

Fig. 2. Overall workflow of the proposed RFI suppression framework. The model is trained using a spatial prior of simulated RFI signals. During inference, the trained model detects RFI and restores the contaminated data in the TFD, which is then transformed back via ISTFT for final imaging.

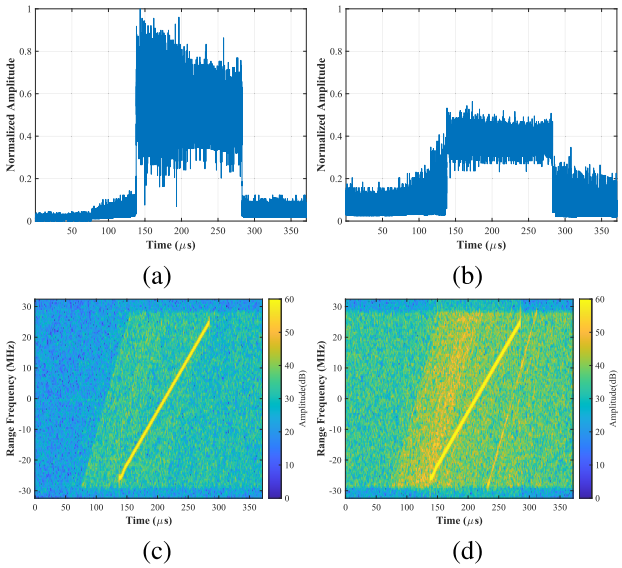

Fig. 3. Illustration of RFI characteristics in dual-polarization TFD from a real Sentinel-1 acquisition. (a) and (b) Time-domain normalized amplitude of the VH and VV channels. (c) and (d) Corresponding TFD. The identical geometric shape and position of the RFI signature (inclined line), despite the difference in the background echo energy between the two channels.

clean data $\mathbf{X}_0$ over T timesteps. The noisy data $\mathbf{X}_t$ in timestep t can be expressed as

$$\mathbf{X}_t = \sqrt{\bar{\alpha}_t}\mathbf{X}_0 + \sqrt{1-\bar{\alpha}_t}\epsilon. \tag{14}$$

To guide the model to focus on restoring the masked region while ensuring the uncontaminated areas remain unchanged,

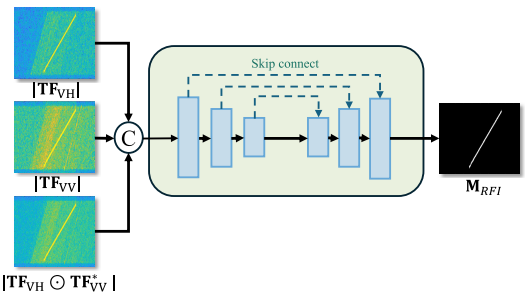

Fig. 4. Function diagram of DPF-UNet. The network takes a three-channel input for segmentation and produces a binary RFI mask $\mathbf{M}_{RFI}$.

we construct a composite tensor $\tilde{\mathbf{X}}_t$

$$\tilde{\mathbf{X}}_t = \mathbf{X}_t \odot \mathbf{M}_{RFI} + \mathbf{X}_0 \odot (1 - \mathbf{M}_{RFI}). \tag{15}$$

This ensures the mask region contains noisy data $\mathbf{X}_t$ and the clean region retains the original clean data $\mathbf{X}_0$. The final input to the denoising model p_θ is a concatenation of $\mathbf{X}_{cond}$ and $\tilde{\mathbf{X}}_t$ along the channel dimension.

2) Training Strategy: As a generative model, the data generated by the conditional diffusion model is guided by the spatial context of surrounding unmasked regions and the underlying data distribution learned from the training dataset. To ensure our model generalizes to diverse and unseen RFI scenarios, we design a spatial prior that integrates the structured characteristics of RFI within TFR into the diffusion model's training process. As analyzed in Section II-A, real-world RFI exhibits specific geometric structures in TFR that are determined by its physical parameters, such as carrier

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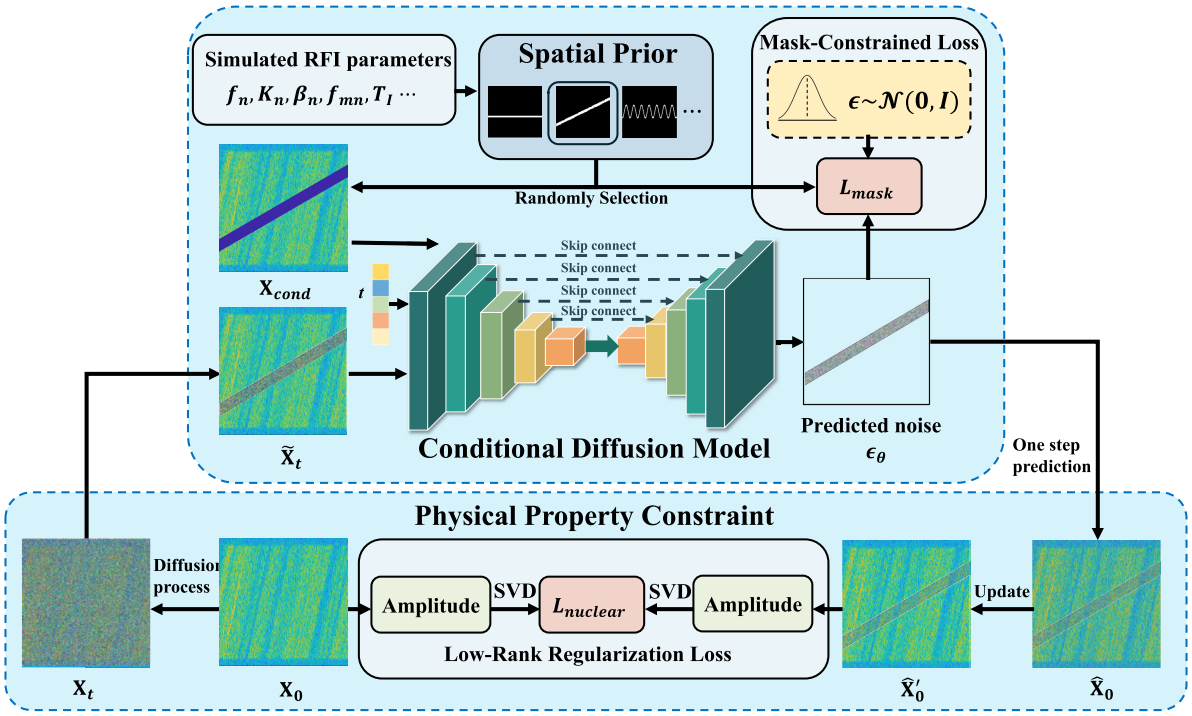

Fig. 5. Overall framework of conditional DDPM training. The conditional diffusion model (top) learns to restore the signal from a noisy input $\tilde{\mathbf{X}}_t$, conditioned on the data $\mathbf{X}_{cond}$. The model is optimized by a composite loss function: a mask-constrained loss Lmask, derived from a spatial prior of simulated RFI, and a low-rank regularization loss L_{nuclear} that enforces physical property constraint via one-step prediction $\hat{\mathbf{X}}_0'$.

frequency f_n, chirp rate K_n, and modulation factor β_n. Training a model on only a few fixed shapes of missing data will severely limit its generalization ability. To this end, we construct a mask library by simulating a wide range of physically realistic RFI parameters. During each training iteration, a mask $\mathbf{M}_{RFI}$ is randomly sampled from this library to construct the conditional data $\mathbf{X}_{cond}$. To ensure balanced learning across different interference types, the mask is sampled following a uniform distribution, ensuring equal probability for each RFI type. This strategy prompts the model to learn a more robust and generalizable spatial context reasoning, enabling it to effectively restore data corrupted by diverse RFI patterns.

3) Composite Loss Function: The standard loss function L_{simple} in diffusion models optimizes the mean square error between the predicted noise and the ground-truth noise. However, for the specific task of SAR TFD restoration, this generic loss has two critical limitations: it lacks explicit constraints on the physical properties of the SAR TFD, and it treats all TFD elements equally, failing to focus the model's learning capacity on the RFI-corrupted regions that require restoration. To overcome these limitations, we further design a composite loss function.

First, to enforce the physical properties of signals, we introduce a low-rank regularization loss L_{nuclear}, based on the fact that the amplitude of a clean SAR TFR is strongly low-rank. Specifically, a preliminary estimate is obtained via one-step prediction using predicted noise, expressed as

$$\hat{\mathbf{X}}_0 = \left(\mathbf{X}_t - \epsilon_\theta \sqrt{1-\bar{\alpha}_t}\right) \Big/ \sqrt{\bar{\alpha}_t}. \tag{16}$$

Then, the estimation is refined into a more reliable estimate $\hat{\mathbf{X}}_0'$ using the spatial prior, expressed as

$$\hat{\mathbf{X}}_0' = \hat{\mathbf{X}}_0 \odot \mathbf{M}_{RFI} + \mathbf{X}_{cond}. \tag{17}$$

Based on this, L_{nuclear} penalizes the difference between the nuclear norm and that of the ground truth

$$L_{\text{nuclear}} = \frac{1}{\sqrt{N}} \left\| \left\| \hat{\mathbf{X}}_0'^{(1)} \right\|_* - \left\| \mathbf{X}_0^{(1)} \right\|_* \right\|_1 \tag{18}$$

where the superscript (1) denotes the magnitude channel of TFR, and N is the total number of matrix elements. The low-rank property of the TFD amplitude stems from the frequency stability of ground echoes, where the energy is predominantly concentrated in the first few singular values [37]. This guides the model to generate results that adhere to the low-rank structure.

Second, to focus the network's attention on the restoration task, we apply a mask constraint to the standard loss, formulating the Lmask term. As defined in (19), this loss computes the noise prediction error only within the regions indicated by the RFI mask $\mathbf{M}_{RFI}$, compelling the model to concentrate its efforts on the inpainting task

$$L_{\text{mask}} = \mathbb{E}_{x_0,t,\epsilon}\left[\|(\epsilon - \epsilon_\theta(\mathbf{X}_t, t)) \odot \mathbf{M}_{RFI}\|^2\right]. \tag{19}$$

The final composite loss function is a weighted sum of these two components, expressed as

$$L = L_{\text{mask}} + \lambda \cdot L_{\text{nuclear}} \tag{20}$$

where λ is a regularization parameter that balances the two loss terms.

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D. Inference Process

Once trained, the model p_θ can restore a new RFI-contaminated TFR $\mathbf{X}_{RFI}$ through an iterative reverse denoising process. The process is conditioned on the uncontaminated regions, defined as

$$\mathbf{X}_{cond} = \mathbf{X}_{RFI} \odot (1 - \mathbf{M}_{RFI}) \tag{21}$$

where $\mathbf{M}_{RFI}$ is the RFI localization resulted from DPF-Unet. Starting from a random Gaussian noise $\mathbf{X}_T \sim \mathcal{N}(0, \boldsymbol{I})$, at each timestep t, the model first estimates the clean image based on the conditional information $\mathbf{X}_{cond}$ and the current noisy sample $\mathbf{X}_t$. This is achieved by first one-step predicting $\hat{\mathbf{X}}_0$ and then updating it to $\hat{\mathbf{X}}_0'$ using the spatial prior, as detailed in (16) and (17). With the refined estimate, the sample for the previous timestep $t-1$ can be expressed as

$$\mathbf{X}_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\hat{\mathbf{X}}_0' + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2}\epsilon_\theta + \sigma_t \epsilon \tag{22}$$

where $\epsilon \sim \mathcal{N}(0, \boldsymbol{I})$ is additional Gaussian noise, and σ_t can be expressed as

$$\sigma_t(\delta) = \delta \sqrt{(1 - \bar{\alpha}_{t-1})/(1 - \bar{\alpha}_t)} \sqrt{1 - \bar{\alpha}_t/\bar{\alpha}_{t-1}} \tag{23}$$

where $\delta \in [0, 1]$ is a hyperparameter that controls the noise level. By repeating the above steps, the RFI-contaminated X_{RFI} is finally restored to a clean TFR X_0 at $t = 0$.

Algorithm 1 Training

Input:
The clean SAR TF data $\mathbf{X}_0$, RFI mask library, and U-Netp_θ .
Output:
U-Net p_θ.
Repeat
$t \sim Uniform(1, \ldots, T)$;
$\epsilon \sim \mathcal{N}(0, \boldsymbol{I})$;
Sample RFI mask $\mathbf{M}_{RFI}$ randomly from the library;
$\mathbf{X}_{cond} = \mathbf{X}_0 \odot (1 - \mathbf{M}_{RFI})$;
Forward diffusion: $\mathbf{X}_t = \sqrt{\bar{\alpha}_t}\mathbf{X}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$;
Construct $\tilde{\mathbf{X}}_t = \mathbf{X}_t \odot \mathbf{M}_{RFI} + \mathbf{X}_0 \odot (1 - \mathbf{M}_{RFI})$;
Compute $\hat{\mathbf{X}}_0 = (\mathbf{X}_t - \epsilon_\theta \sqrt{1 - \bar{\alpha}_t})/\sqrt{\bar{\alpha}_t}$;
Compute $\hat{\mathbf{X}}_0' = \hat{\mathbf{X}}_0 \odot \mathbf{M}_{RFI} + \mathbf{X}_{cond}$;
Compute $L_{nuclear}$ by (18);
Compute L_{mask} by (19);
Joint loss: $L \leftarrow L_{mask} + \lambda \cdot L_{nuclear}$;
Take gradient descent ∇_θ;
Update p_θ;
Until converged or max epoch reached

The detailed training and inference processes are also shown in Algorithms 1 and 2, respectively.

Algorithm 2 Inference

Input:
SAR TF data with RFI $\mathbf{X}_{RFI}$, RFI localization result $\mathbf{M}_{RFI}$, and U-Net p_θ.
Output:
Restored SAR TF data $\mathbf{X}_0$.
Initialize
$\mathbf{X}_{cond} = \mathbf{X}_{RFI} \odot (1 - \mathbf{M}_{RFI})$;
Sample $\mathbf{X}_T \sim \mathcal{N}(0, \boldsymbol{I})$;
for $t = T, \ldots, 1$ **do**
$\hat{\mathbf{X}}_0 = (\mathbf{X}_t - \epsilon_\theta \sqrt{1 - \bar{\alpha}_t})/\sqrt{\bar{\alpha}_t}$;
$\hat{\mathbf{X}}_0' = \hat{\mathbf{X}}_0 \odot \mathbf{M}_{RFI} + \mathbf{X}_{cond}$;
$\mathbf{X}_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\hat{\mathbf{X}}_0' + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2}\epsilon_\theta + \sigma_t \epsilon$;
end for

IV. Experimental Results

In this section, a series of experiments were conducted to validate the effectiveness of our method. The datasets were constructed from Sentinel-1 dual-polarization, Level-0 raw data. The method was evaluated on both clean data with simulated RFI and on data with real RFI contamination. To assess the performance, we conducted comparative experiments from two perspectives. Specifically, we compared the DPF-UNet with the classical TFN [21] and constant false alarm rate (CFAR) to evaluate the RFI detection and localization capabilities. Subsequently, the overall suppression performance was compared with several classical methods, FNF, TFN, RPCA [25], and recent deep learning-based methods IMN [35] and PISNet [37]. These comparative methods are all implemented in the TFD.

A. Dataset Description

This section describes the datasets used in our experiments. We primarily use real-world data from the Sentinel-1 satellite in interferometric wide (IW) mode. The selected scenes cover Chungcheong, Korea, as shown in Fig. 6(a). The size of each raw echo is $1400 \times 19\,798$. The specific datasets are categorized as follows.

1) *Data 1 (RFI-Free):* Acquisition ID S1A_IW_RAW__0SDV_20200328T093137_20200328T093209_031874_03ADE0_7A40. This dataset is not contaminated by RFI. It is first used to build a clean training set for the diffusion model to learn the general physical features of SAR signals. Then, by adding simulated interference to this data, we quantitatively evaluate the method's performance and its generalization ability to unseen RFI patterns and different geographical areas.
2) *Data 2 (Real RFI):* Acquisition ID S1A_IW_RAW__0SDV_20200409T093137_20200409T093209_032049_03B411_8B4B and S1B_IW_RAW__0SDV_20200221T170025_20200221T170058_020370_026955_E1FB. These datasets contain real RFI and are used to evaluate the method's effectiveness in complex, real-world electromagnetic environments.
3) *Data 3 (RFI-Free):* RADARSAT-1 data used to verify the cross-platform generalization of the proposed method.

B. Evaluation Metrics

To comprehensively assess the performance of the proposed method, we utilize two sets of metrics: one for evaluating the RFI detection accuracy of DPF-UNet, and the other

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for assessing the final RFI suppression performance in both simulated and real RFI scenarios.

1) Evaluation Metrics for RFI Detection: To evaluate the performance of the proposed DPF-UNet under various SIRs, we employ metrics including precision, recall, $F1$-score, and mean intersection over union (mIoU). These metrics are defined as

$$\text{Precision} = \frac{TP}{TP + FP} \tag{24}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{25}$$

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{26}$$

and

$$\text{mIoU} = \frac{1}{N} \sum_{i=1}^{N} \text{IoU}_i \tag{27}$$

where TP, FP, and FN denote true positive, false positive, and false negative, respectively, and IoU_i is the intersection over union for the ith class.

2) Evaluation Metrics for RFI Suppression in Simulated Experiments: For the performance evaluation of the proposed method on the simulated experiments, we selected several indicators. First, signal distortion ratio (SDR) [37] is defined as

$$\text{SDR} = 10 \log_{10} \left(\frac{\sum \|\hat{x} - x\|^2}{\sum \|x\|^2} \right) \tag{28}$$

where x and $\hat{x}$ are the amplitudes of the original RFI-free echo and RFI-suppressed echo, respectively. SDR reflects the removal of interference and the loss caused by recovery. A lower SDR indicates better performance.

Second, structural similarity (SSIM) [55] and root-mean-square error (RMSE) are used to evaluate the similarity and accuracy of the restored details in the TF domain. SSIM is defined as

$$\text{SSIM} = \left[l(m,n)^{\alpha} \right] \left[c(m,n)^{\beta} \right] \left[s(m,n)^{\gamma} \right] \tag{29}$$

where $l(m,n)$, $c(m,n)$, and $s(m,n)$ are brightness, contrast, and structure comparison, respectively. In addition, RMSE is defined as

$$\text{RMSE} = \sqrt{\frac{1}{MN} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} \left(\mathbf{X}(m,n) - \hat{\mathbf{X}}(m,n) \right)^2} \tag{30}$$

where $\mathbf{X}(m,n)$ and $\hat{\mathbf{X}}(m,n)$ are the value of the pixel at position (m,n) in the original and restored TFR, respectively.

Meanwhile, we also introduce another indicator, peak signal-to-noise ratio (PSNR), to evaluate the quality of the focused SAR image, which is defined as

$$\text{PSNR} = 10 \log_{10} \left(\frac{max \|\mathbf{X}\|^2}{\text{MSE}} \right) \tag{31}$$

where MSE is the mean square error between $\hat{\mathbf{X}}$ and $\mathbf{X}$. It should be noted that all quantitative metrics for TFD restoration are calculated based on linear amplitude data rather than on a logarithmic scale.

TABLE I
PARAMETERS OF SIMULATED RFI SIGNALS

Type	Parameters	Values
NBI	Number	$\{1, 2, 3\}$
	Carrier Frequency f_n	5.405 GHz $+\mathcal{U}(-23, 23)$MHz
	SINR	$\{-10, -5\}$ dB
LFM WBI	Carrier Frequency f_n	5.405 GHz ±5MHz
	Bandwidth B_n	$\{10, 20, 30\}$ MHz
	SINR	$\{-10, -5\}$ dB
SM WBI	Carrier Frequency f_n	5.405 GHz ±5MHz
	Modulation Factor β_n	$\{30, 35, 40\}\pi$
	Modulation Frequency f_{mn}	$\{20, 30, 40\}$ kHz
	SINR	$\{-10, -5\}$ dB
Mix 1	NBI and SM WBI. Parameters as defined above	
Mix 2	LFM WBI and SM WBI. Parameters as defined above	

3) Evaluation Metrics for RFI Suppression Experiments in Measured Data: For the experiment on measured data with real-world RFI, since the clean ground-truth image is unavailable, we selected three no-reference image quality metrics: average gradient (AG) [37], which reflects image sharpness and texture detail; gray entropy (GE) [46], which measures the richness of information in the image; mean square deviation (MSD) [43], which reflects the image contrast. Higher AG and MSD values and lower GE values generally indicate better image quality and information content. The definitions are as

$$\text{AG} = \frac{1}{(M-1)(N-1)} \sum_{m=1}^{M-1} \sum_{n=1}^{N-1} \frac{1}{4} \times \sqrt{\left(\frac{\partial \mathbf{X}(m,n)}{\partial m} \right)^2 + \left(\frac{\partial \mathbf{X}(m,n)}{\partial n} \right)^2} \tag{32}$$

$$\text{GE} = -\sum_{i=0}^{255} p_i \log p_i \tag{33}$$

and

$$\text{MSD} = \frac{\|\mathbf{X} - \text{mean}(\mathbf{X})\|_F}{MN} \tag{34}$$

where $\partial \mathbf{X}(m,n)/\partial m$ and $\partial \mathbf{X}(m,n)/\partial n$ are the first-order differences in the horizontal and vertical directions of image, and p_i is the probability of SAR image grayscale values, with $i \in [0, 255]$ representing linearly scaled 8-bit amplitude values.

C. Training Details

All experiments were conducted on a workstation equipped with an Intel Xeon Gold 6326 CPU, an NVIDIA RTX 4090 GPU (24 GB), and 128 GB of RAM, and the Ubuntu 20.04 operating system. The training of the DPF-UNet and conditional diffusion model was implemented in PyTorch 2.5.1.

For DPF-UNet, the training set was generated by adding simulated RFI to the clean, dual-channel data 1. The RFI parameters are detailed in Table I. To simulate the polarization difference of RFI across different channels, the amplitude ratio A_{VH}/A_{VV} is modeled as a random variable sampled from

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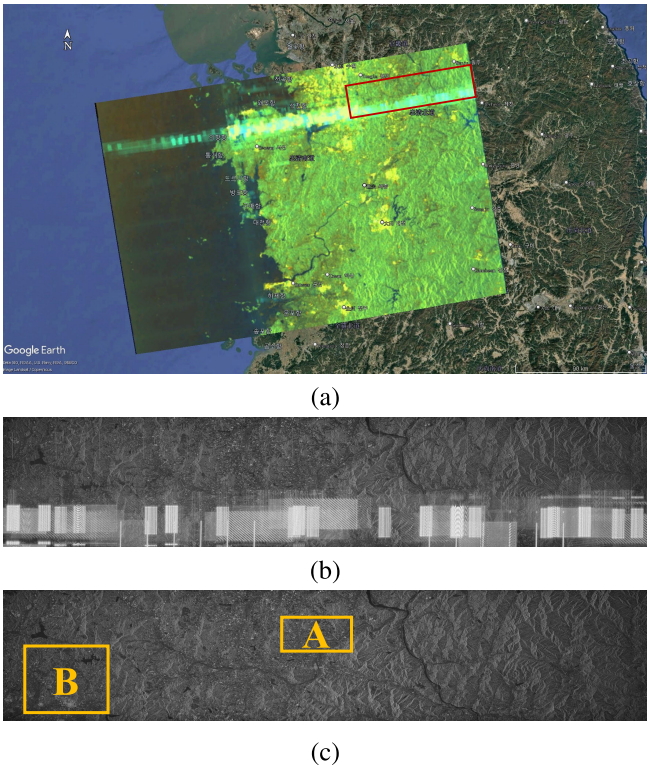

Fig. 6. Descriptions of experimental data. (a) Geolocations of the Sentinel-1 data. (b) Amplitude image of the selected region with RFI. (c) Amplitude image of the selected region without RFI. The regions A and B shown in (c) are chosen for detailed analysis and quantitative evaluation.

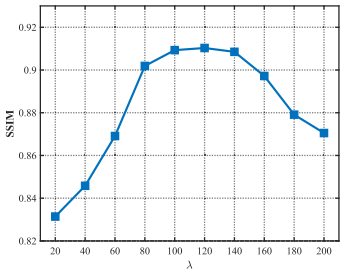

Fig. 7. Sensitivity analysis of the hyperparameter λ.

the interval [0.3, 3]. Meanwhile, we used a composite loss function of cross-entropy and Dice loss.

For the conditional diffusion model, the training dataset comprises 7000 TFD samples derived from clean data 1. Specifically, the raw data was transformed into TFDs pulse by pulse using a Hann window (length 128, overlap 96), and each TFD was divided into 128 × 128 blocks. The RFI mask library was generated using the parameters in Table I. These parameters were selected to balance mask richness with training efficiency, ensuring diverse RFI patterns and maximizing spatial randomness. Consequently, by randomly sampling from these specified ranges, the model achieves robust generalization capabilities. The diffusion model follows the standard U-Net architecture proposed in [56], utilizing ResNet blocks and attention mechanisms. The U-Net architecture consists of an encoder, a MidBlock, and a decoder, as shown in Fig. 8. First, the input data is mapped to 128 channels via a 3 × 3 convolutional layer with a stride and

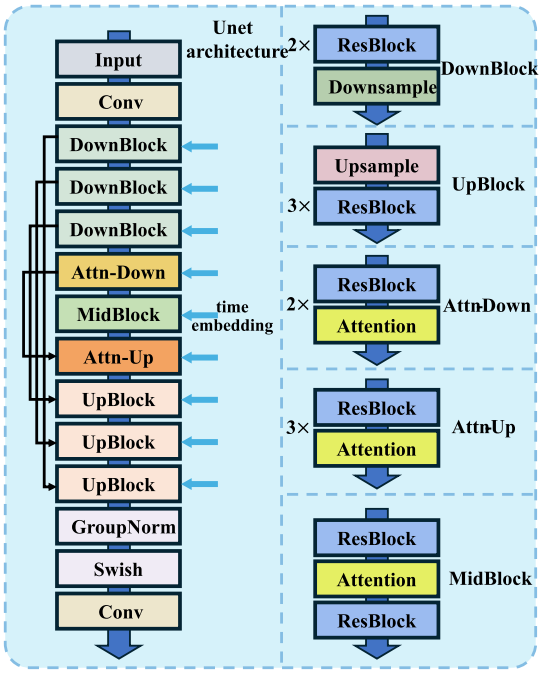

Fig. 8. Detailed architecture of the U-Net model [56]. The left shows the overall structure (black arrows: skip connections; blue arrows: time embeddings). The right details the internal block compositions. Attention mechanisms are exclusively deployed at the lowest resolution (Attn-Down, MidBlock, and Attn-Up).

padding of 1. The encoder employs a sequence of DownBlocks to reduce spatial resolution, with channel dimensions specified as 128, 128, 256, and 256. Attention modules are added after the ResNet modules when the feature map resolution reaches 16. The MidBlock bridges the encoder and decoder and incorporates an attention block. The decoder then restores the spatial resolution via Attn-Up and UpBlocks. Feature maps from the encoder are concatenated with corresponding decoder features via multiscale skip connections. Finally, the network generates a two-channel output through group normalization, a swish activation, and a final 3 × 3 convolutional layer. The ResNet blocks adopt the standard architecture commonly used in [40]. Specifically, it integrates the time-step embedding into the feature maps via a linear projection and element-wise addition following the first convolutional layer. For the attention mechanism, we utilize the standard scaled dot-product attention [57].

The model was trained using the Adam optimizer with a learning rate of $2e^{-5}$ and a batch size of 16 for 800 epochs, with timestep T set to 500. The noise level parameter δ controlling the stochasticity of the reverse process is set to 0. Furthermore, in order to select a suitable hyperparameter λ, a series of experiments was conducted with λ varying from 20 to 200. The SSIM curve of the restored TFR is illustrated in Fig. 7. Based on the results, we selected $\lambda = 120$ as the optimal setting due to its robust performance.

D. Performance of RFI Detection of DPF-UNet

To validate the effectiveness of the proposed DPF-UNet for RFI detection, a test dataset independent of the training set was constructed. This dataset spans an interference-to-signal ratio

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TABLE II
QUANTITATIVE EVALUATION OF RFI DETECTION PERFORMANCE IN TEST SET

	Precision	Recall	F1-score	mIoU
TFN	0.6926	0.3540	0.4686	0.6326
CFAR	0.9061	0.2921	0.4418	0.6230
Proposed	0.9608	0.9393	0.9499	0.9497

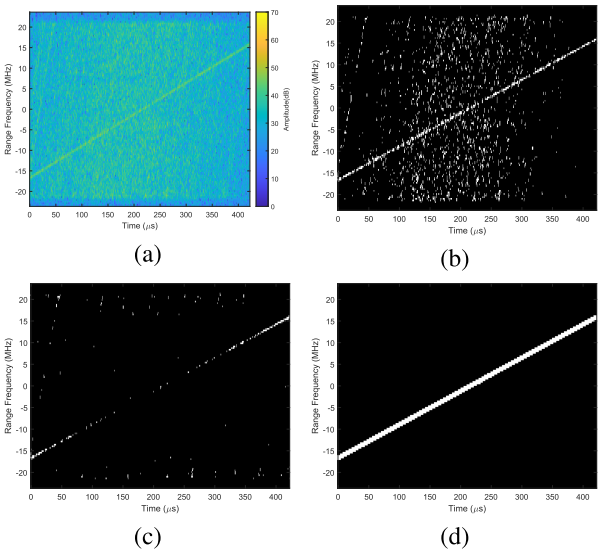

Fig. 9. Example of RFI detection results under a weak interference scenario (ISR = −12 dB). (a) RFI-contaminated TFD. (b) TFN. (c) CFAR. (d) DPF-UNet.

(ISR) range from −15 to 15 dB. Notably, the RFI simulation parameters in the test set differ from the training configuration listed in Table I. Table II summarizes the quantitative evaluation metrics for the test set. The results demonstrate that the DPF-UNet achieves substantial improvements in precision, recall, $F1$-score, and mIoU compared to the conventional TFN method and CFAR detector, demonstrating its superior detection performance.

Fig. 9 illustrates a representative example of weak RFI detection at an ISR of −12 dB. In such low ISR scenarios, the RFI signal energy is lower than that of strong useful signals within the scene. Consequently, the TFN method, which relies on fixed amplitude thresholds, fails to differentiate between weak interference and clean echoes, leading to significant false alarms. While the CFAR detector preserves relatively high precision, it often yields a fragmented morphology when detecting weak RFI with continuous geometric structures, resulting in missed detections. In contrast, the DPF-UNet accurately localizes and detects weak RFI signals. This superior performance stems from the network's ability to overcome the limitations of simple intensity-based detection by integrating dual-polarization data. By leveraging the high cross-channel coherence and distinct geometric structures of RFI in TFD, the model extracts deep discriminative features. These features enable the precise localization and extraction of RFI signals based on specific texture patterns, even when they are submerged within strong echoes.

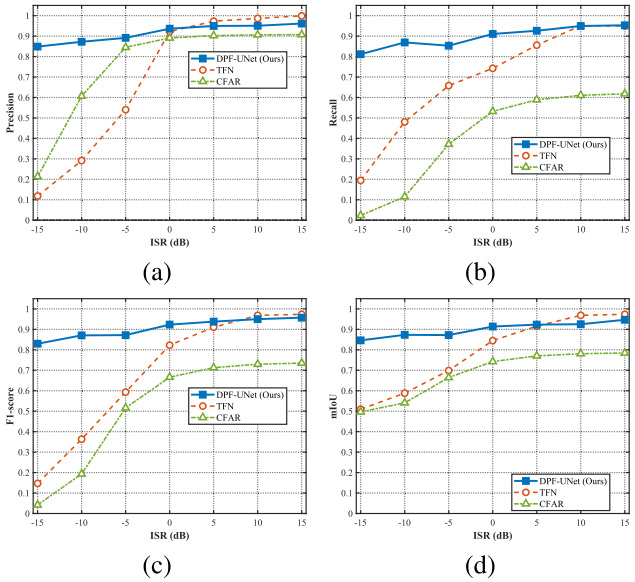

Fig. 10. Quantitative comparison of detection performance across different ISR levels. (a) Precision. (b) Recall. (c) $F1$-score. (d) mIoU.

Furthermore, to assess the robustness of the proposed method across varying interference intensities, the test data were grouped by ISR levels ranging from −15 to 15 dB. The evaluation metrics as a function of ISR are illustrated in Fig. 10(a)–(d). The resulting curves indicate that under high ISR conditions, the conventional TFN method yields acceptable performance due to the distinct power contrast between the RFI and the background. However, as the ISR decreases, the performance of TFN and CFAR degrades rapidly, leading to a significant increase in both missed detections and false alarms. In contrast, the performance of the DPF-UNet remains highly stable across the entire dynamic range, maintaining superior detection accuracy even under extremely low ISR conditions.

E. Performance of RFI Suppression in Simulated Experiments

1) Evaluation on TFD: We first evaluate the performance of the proposed method on clean data added with various types of simulated RFI as defined in Table I. Fig. 11 presents a visual comparison of the suppression results in TFD for five representative RFI patterns, with the corresponding quantitative metrics listed in Table III. The experimental results of TFD are displayed in linear amplitude.

The results show that although the classical notch filter methods FNF and TFN demonstrate satisfactory interference removal in the case of NBI, their performance deteriorates significantly against wideband or mixed RFI scenarios. This degradation occurs because, as the RFI bandwidth increases, a larger portion of useful signals is inevitably removed along with the interference, leading to a loss of information and poor image quality.

As a representative low-rank and sparse decomposition approach, RPCA can effectively suppress NBI. However, it fails to achieve satisfactory performance when addressing

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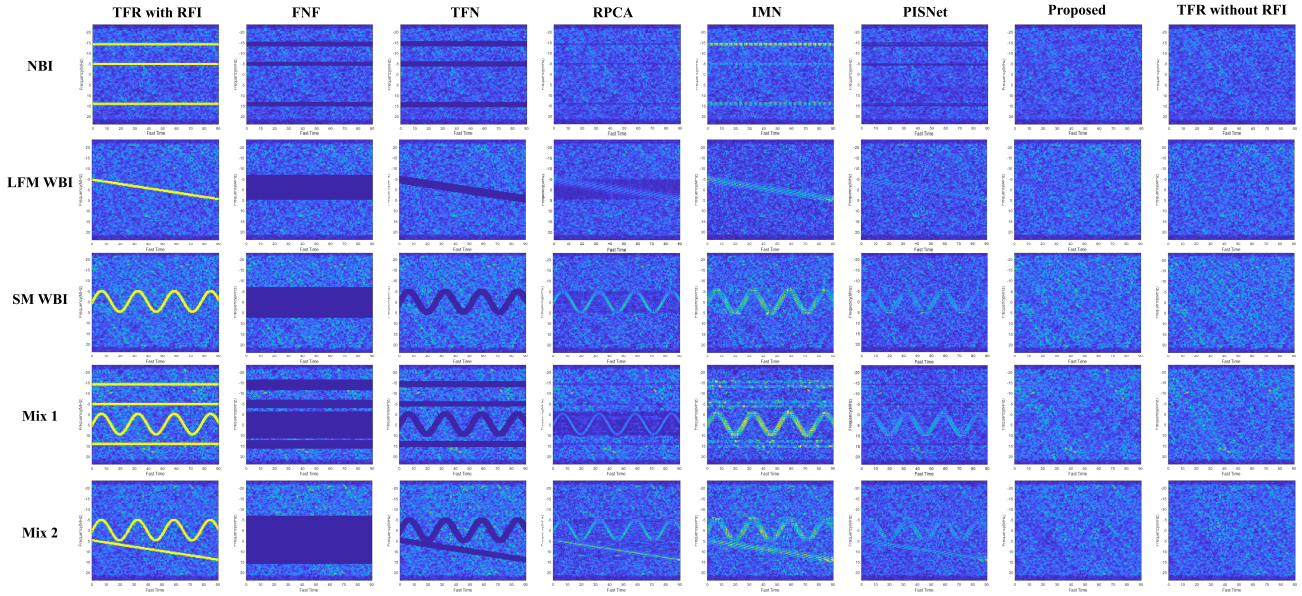

Fig. 11. Comparisons of RFI suppression results of different methods in the case of five representative RFI patterns.

TABLE III

QUANTITATIVE EVALUATION OF RFI SUPPRESSION PERFORMANCE IN TFD. ↑ MEANS HIGHER VALUES ARE BETTER. ↓ MEANS LOWER VALUES ARE BETTER. Δ REPRESENTS THE AMOUNT OF CHANGE OF THE PROPOSED METHOD COMPARED TO THE SUBOPTIMAL

		With RFI	FNF	TFN	RPCA	IMN [35]	PISNet [37]	Proposed	Δ
NBI	SDR(dB)↓	-6.04896	-11.20350	-14.96757	-15.82552	-10.54364	-14.52479	**-16.49773**	**-0.67221**
	SSIM↑	0.13677	0.83051	0.80074	0.78631	0.57404	0.83156	**0.91697**	**+0.08541**
	RMSE↓	3.22913	0.39920	0.44025	0.29838	0.73495	0.33272	**0.20162**	**-0.09676**
LFM WBI	SDR(dB)↓	-8.80601	-13.95700	-12.60703	-14.91188	-12.97566	-13.93434	**-16.32107**	**-1.40919**
	SSIM↑	0.14793	0.73563	0.85187	0.53197	0.28841	0.59607	**0.90955**	**+0.05768**
	RMSE↓	1.85703	0.52193	0.29363	0.43488	0.45085	0.29147	**0.24354**	**-0.04793**
SM WBI	SDR(dB)↓	-8.74068	-14.18408	-12.35108	-13.89510	-14.08787	-14.15971	**-17.24591**	**-3.06183**
	SSIM↑	0.10152	0.68176	0.83832	0.66054	0.47239	0.68540	**0.88328**	**+0.04496**
	RMSE↓	1.83834	0.58212	0.31740	0.43623	0.45834	0.33695	**0.24982**	**-0.06758**
Mix 1	SDR(dB)↓	-5.04870	-13.06503	-13.85070	-14.49285	-13.45000	-11.43170	**-15.21449**	**-0.72164**
	SSIM↑	0.11279	0.40498	0.67024	0.71353	0.47230	0.72883	**0.84219**	**+0.11336**
	RMSE↓	3.85752	0.76470	0.54042	0.39881	0.69950	0.39130	**0.32274**	**-0.06856**
Mix 2	SDR(dB)↓	-7.65085	-13.11836	-14.11540	-13.05533	-13.79388	-11.84646	**-15.46605**	**-1.35065**
	SSIM↑	0.12011	0.50289	0.71702	0.39029	0.52963	0.72114	**0.80432**	**+0.08318**
	RMSE↓	2.50831	0.71736	0.41135	0.53817	0.51987	0.36904	**0.28227**	**-0.08677**

WBI or mixed RFI. This failure stems from the fact that complex WBI and mixed RFI do not conform to the strict low-rank assumption, leading to residual RFI components in the reconstructed region. Furthermore, the uncontaminated regions may also be degraded by the low-rank approximation, making it difficult to accurately recover these useful signals.

Compared with traditional approaches, deep learning-based methods offer a better tradeoff between RFI removal and the preservation of useful signals. However, our experimental results indicate that IMN appears vulnerable to diverse RFI types, leaving significant residual RFI, especially under low-SINR conditions. PISNet, which incorporates a low-rank prior, performs better in certain interference scenarios and attains suboptimal results in several evaluation metrics. Nevertheless, its end-to-end architecture remains sensitive to complex RFI cases, resulting in both residual RFI and a noticeable loss of useful signal information.

In contrast, our proposed method consistently achieves the best performance in TFD. As shown in the seventh column of Fig. 11, it not only effectively removes RFI but also fully preserves the uncontaminated regions with high fidelity. In contrast to traditional notch filtering methods, our generative approach restores the RFI-corrupted areas, ensuring textural and structural continuity with the surrounding data. This superiority is reflected in the quantitative metrics in Table III, where our method outperforms all other methods across each scenario.

2) Evaluation on Focused SAR Images: To evaluate the performance of our method on the final focused SAR images, we designed two challenging simulated scenarios. Scenario I was designed to test the performance against a single wide-band interference type, where we added an LFM RFI with a bandwidth of 30 MHz. Scenario II then evaluated the performance in a more complex situation by simulating a

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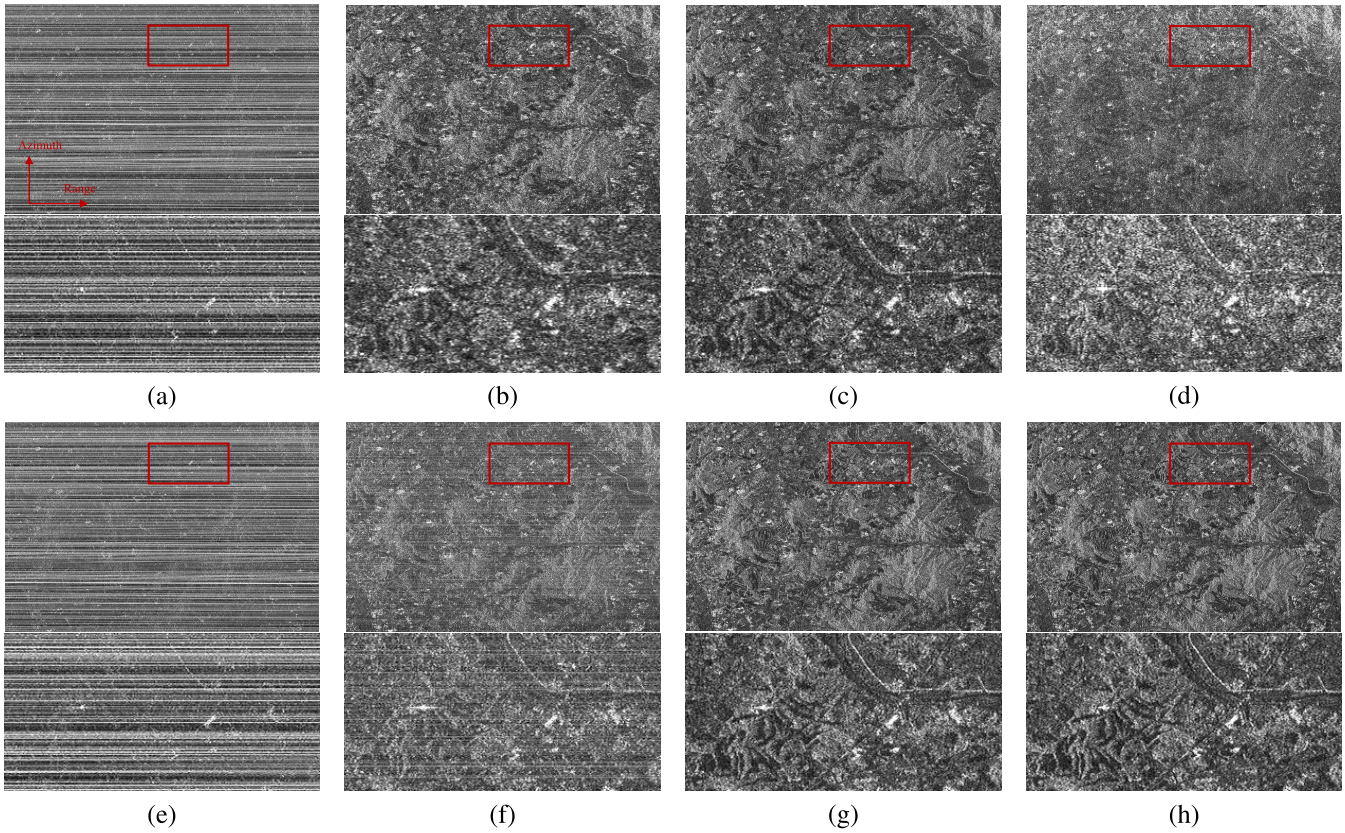

Fig. 12. Imaging results of selected region A in Scenario I. The top row shows the selected regions, while the bottom row displays the corresponding ROIs from the areas indicated by the red boxes. (a) RFI-contaminated image. (b) FNF. (c) TFN. (d) RPCA. (e) IMN. (f) PISNet. (g) Proposed method. (h) Ground truth image without RFI.

TABLE IV

QUANTITATIVE EVALUATION OF RFI SUPPRESSION PERFORMANCE IN SAR IMAGE. ↑ MEANS HIGHER VALUES ARE BETTER. ↓ MEANS LOWER VALUES ARE BETTER. **Δ** REPRESENTS THE AMOUNT OF CHANGE OF THE PROPOSED METHOD COMPARED TO THE SUBOPTIMAL

		With RFI	FNF	TFN	RPCA	IMN [35]	PISNet [37]	Proposed	**Δ**
Scenario I	SSIM↑	0.06566	0.47188	0.70991	0.37970	0.08741	0.32004	**0.84183**	**+0.13192**
	PSNR (dB)↑	11.57355	17.72366	19.70875	14.04192	12.02764	16.31814	**21.95979**	**+2.25104**
	RMSE↓	0.62558	0.38795	0.30869	0.47082	0.59371	0.36229	**0.13396**	**-0.17473**
Scenario II	SSIM↑	0.04244	0.68118	0.72890	0.48061	0.07224	0.11788	**0.87215**	**+0.14325**
	PSNR (dB)↑	12.79251	19.44648	20.16590	19.43965	11.42884	14.14456	**21.56102**	**+1.39512**
	RMSE↓	0.65689	0.38441	0.23024	0.30559	0.76856	0.56220	**0.16946**	**-0.06078**

composite interference environment, combining an LFM RFI (bandwidth B_n = 30 MHz) with an SM RFI (modulation frequency f_{mn} = 50 kHz, modulation factor β_n = 100). For both scenarios, SINR was set to −10 dB. The qualitative and quantitative evaluation results for these two scenarios are presented in Figs. 12 and 13 and Table IV, respectively. It should be noted that the interference cases for these scenarios were selected from our spatial prior's RFI library.

In both simulated scenarios, traditional methods exhibit significant limitations. The core shortcoming of notch filtering-based methods, FNF and TFN, lies in useful signal loss. As shown in Fig. 12(b) and (c), to suppress wideband interference, the filters must remove a substantial portion of useful signals, which inevitably leads to severe degradation of image texture and an elevation of sidelobes. The RPCA method, on the other hand, is limited by its underlying assumptions. When faced with the complex, wideband, and mixed interference in Scenarios I and II, the low-rank property no longer strictly satisfies. Consequently, RPCA fails to effectively separate the interference, leaving a large amount of residual artifacts in the image, as depicted in Figs. 12(d) and 13(d).

The deep learning models IMN and PISNet show a lack of generalization capability in these two complex scenarios. As illustrated in Figs. 12(e) and (f) and 13(e) and (f), these models failed to completely suppress the complex interference, producing severe artifacts or residuals. More importantly, as demonstrated in Scenario II, their end-to-end mapping mechanism can sometimes incorrectly reconstruct the signals, leading to blurred details in areas such as buildings.

In contrast, our proposed method demonstrates optimal performance in all scenarios. By reframing the problem as a probabilistic conditional generation task, DDPM can generate

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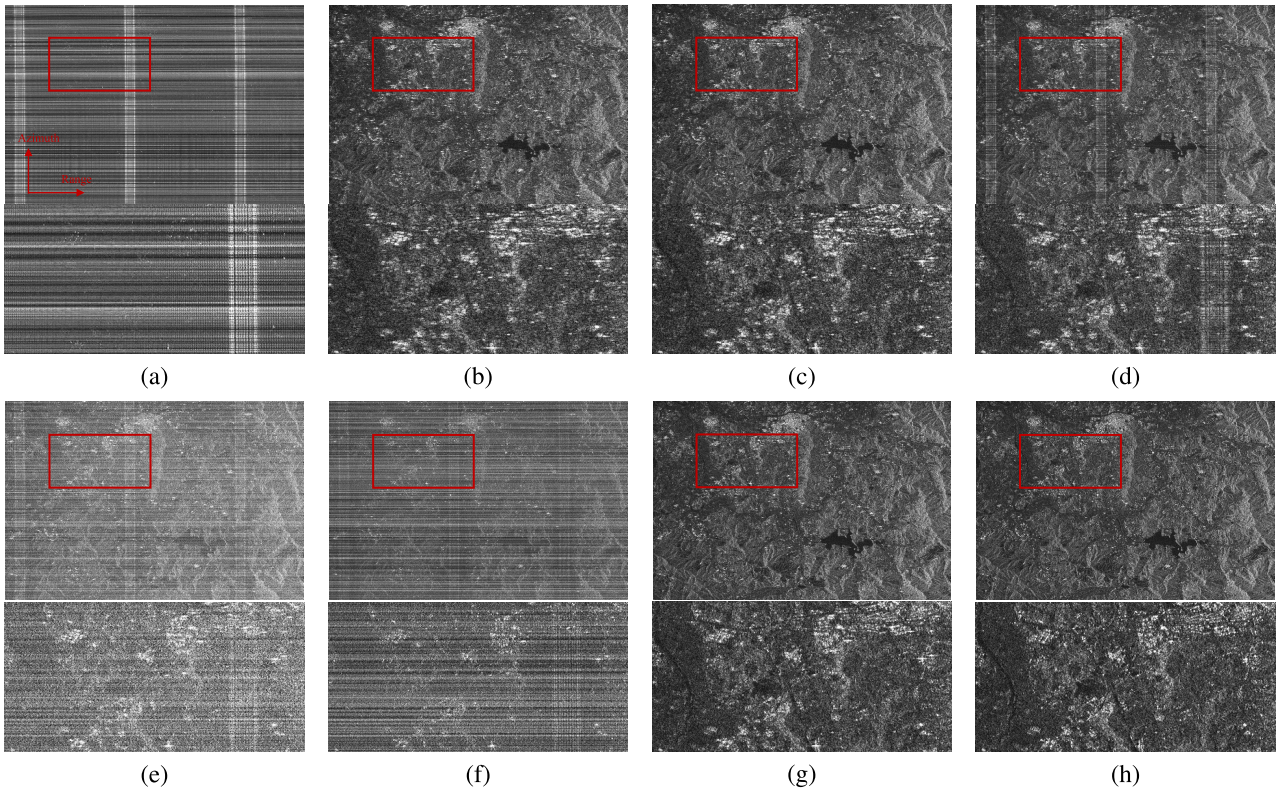

Fig. 13. Imaging results of selected region B in Scenario II. The top row shows the selected regions, while the bottom row displays the corresponding ROIs from the areas indicated by the red boxes. (a) RFI-contaminated image. (b) FNF. (c) TFN. (d) RPCA. (e) IMN. (f) PISNet. (g) Proposed method. (h) Ground truth image without RFI.

TABLE V

QUANTITATIVE EVALUATION OF GENERALIZATION PERFORMANCE. ↑ MEANS HIGHER VALUES ARE BETTER. ↓ MEANS LOWER VALUES ARE BETTER. Δ REPRESENTS THE AMOUNT OF CHANGE OF THE PROPOSED METHOD COMPARED TO THE SUBOPTIMAL. SDR IS EXPRESSED IN DB; SSIM IS DIMENSIONLESS; AND RMSE REPRESENTS THE ERROR IN LINEAR AMPLITUDE

		With RFI	FNF	TFN	RPCA	IMN [35]	PISNet [37]	Proposed	Δ
Unseen RFI	SDR(dB)↓	-8.64124	-11.26159	-11.67509	-12.29665	-13.57697	-12.50740	**-14.80786**	**-1.23089**
	SSIM↑	0.12295	0.17782	0.84398	0.15036	0.31647	0.54650	**0.91293**	**+0.06895**
	RMSE↓	1.87755	0.93766	0.27943	0.73910	0.62062	0.42396	**0.24835**	**-0.03108**
Different Region	SSIM↑	0.03418	0.31551	0.42017	0.31304	0.17126	0.11470	**0.75075**	**+0.33058**
	PSNR(dB)↑	12.45372	16.39218	17.74995	16.39987	14.21437	13.67749	**19.29438**	**+1.54443**
	RMSE↓	0.59583	0.37861	0.32382	0.37828	0.48650	0.51752	**0.12417**	**-0.19965**
Different Platform	SSIM↑	0.19331	0.24424	0.31174	0.33701	0.12664	0.12878	**0.66595**	**+0.32894**
	PSNR(dB)↑	11.99096	12.77152	13.46223	13.52068	11.43524	11.53754	**20.11143**	**+6.59074**
	RMSE↓	0.62015	0.56685	0.52352	0.52001	0.66112	0.65338	**0.13585**	**-0.38416**

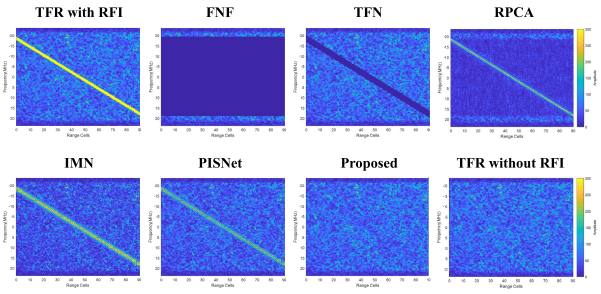

Fig. 14. Comparisons of suppression performance for different methods on an unseen RFI pattern. TFD is contaminated with an LFM WBI with a different chirp rate not included in the training set.

the signal for the contaminated regions with high fidelity, based on the learned true data distribution. As shown in Figs. 12(g) and 13(g), the images have clear details and natural textures, visually appearing closest to the RFI-free ground truth images. The quantitative results in Table IV also demonstrate this visual advantage, with our method significantly leading in all evaluation metrics.

F. Generalization Evaluation

To comprehensively evaluate the generalization capabilities of our proposed framework, we further conducted three experiments below. The first experiment assesses the model's ability to restore unseen RFI patterns, demonstrating that its generalization is not constrained by the spatial priors during training. The second experiment evaluates whether the model maintains effective suppression performance even on data acquired from geographically distinct regions. The third

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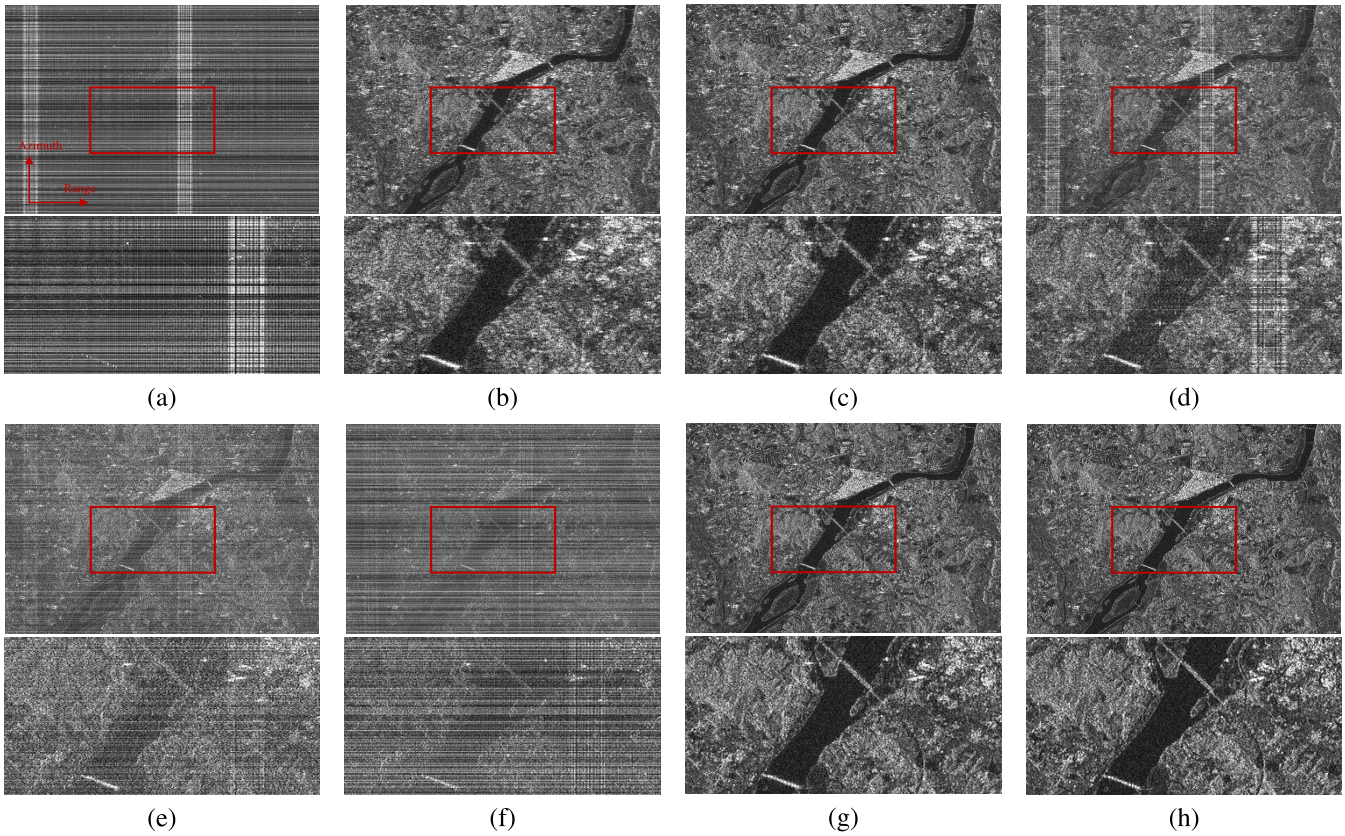

Fig. 15. Imaging results of selected region in different regions. The top row shows the selected regions, while the bottom row displays the corresponding ROIs from the areas indicated by the red boxes. (a) RFI-contaminated image. (b) FNF. (c) TFN. (d) RPCA. (e) IMN. (f) PISNet. (g) Proposed method. (h) Ground truth image without RFI.

experiment proves the model's robustness performance on different SAR platforms. The quantitative performance metrics are detailed in Table V.

1) Generalization to Unseen RFI Patterns: The same model trained on simulated data was employed for this evaluation. Meanwhile, we introduced a new, complex RFI type that is not included in the training set. This new type of interference is LFM RFI with a chirp rate of 418 GHz/s. This unseen interference was added to the clean data, and the suppression performance of each method was evaluated in TFD. Qualitative results are presented in Fig. 14.

As illustrated in Fig. 14, all compared methods performed poorly when confronted with this unseen RFI type. Filtering-based approaches, FNF and TFN, induced significant signal loss due to the over-removal of useful data caused by the wideband interference. RPCA proved ineffective since the complex RFI structure violated its low-rank assumption. Deep learning-based models IMN and PISNet also failed to adequately suppress the complex interference, leaving substantial residual RFI artifacts.

In contrast, the proposed method demonstrated superior generalization performance. Despite the novelty of the RFI pattern, our method reframes the problem as a universal conditional restoration task. The diffusion model leverages its strong generative capability to infer physically reliable reconstructions within contaminated regions based on contextual signal information, without having memorized specific RFI types. Quantitative results in Table V further confirm the superiority of the proposed method, which outperforms all compared methods across evaluation metrics.

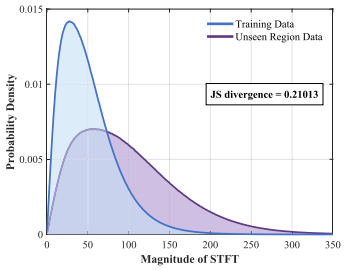

Fig. 16. PDFs of the training data and unseen data.

2) Generalization to Unseen Regions: To verify that the model does not depend exclusively on the specific scene, we tested its generalization capabilities on a dataset from a geographically distinct region outside the training area. Specifically, we selected a clean Sentinel-1 dataset acquired with the same IW imaging mode but from a region geographically distant from the training set, ensuring no spatial overlap. Then, we added the simulated Mix 1 RFI type. Unlike the training data, which is primarily composed of plains and villages, the data in the unseen region consists of urban agglomerations and mountainous areas. Final focused SAR images and

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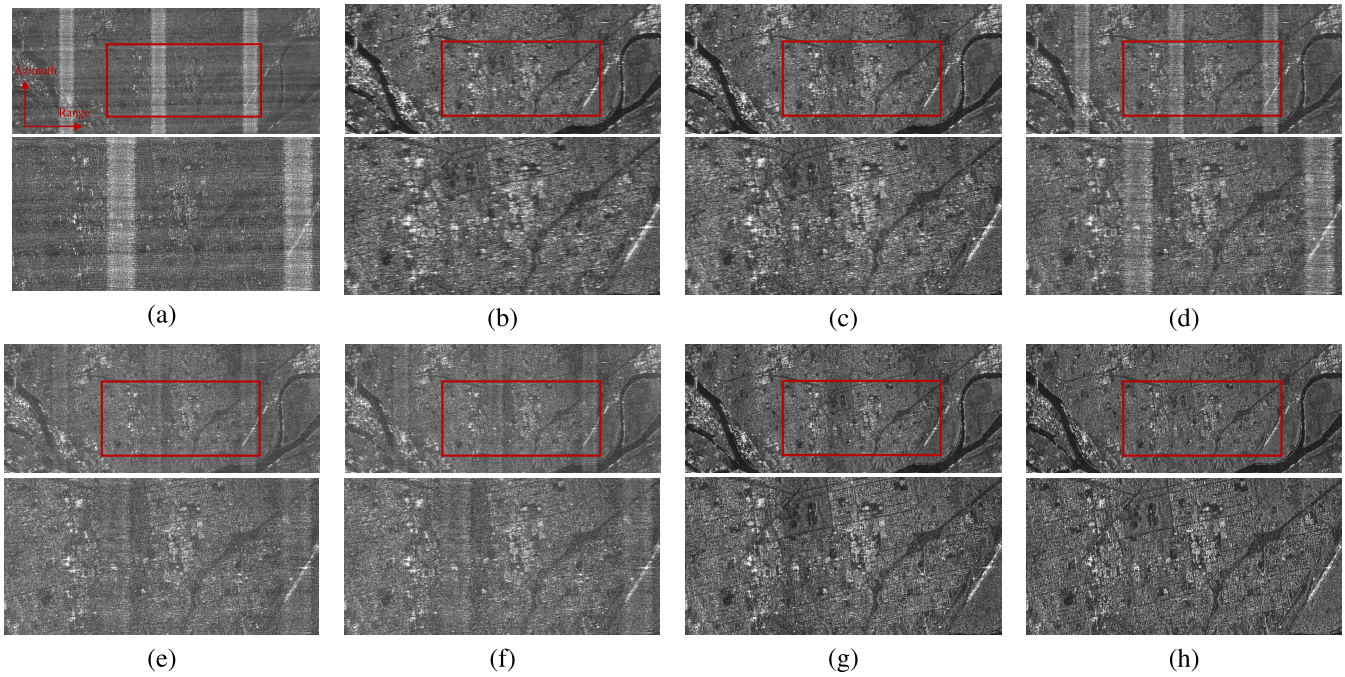

Fig. 17. Imaging results of selected region in Radarsat-1 data. The top row shows the selected regions, while the bottom row displays the corresponding ROIs from the areas indicated by the red boxes. (a) RFI-contaminated image. (b) FNF. (c) TFN. (d) RPCA. (e) IMN. (f) PISNet. (g) Proposed method. (h) Ground truth image without RFI.

suppression results are shown in Fig. 15. The PDFs are shown in Fig. 16 to illustrate the statistical differences between the two regions. The Jensen–Shannon (JS) divergence is 0.21013, indicating a significant difference in the distributions. Notably, the same trained model was applied without any fine-tuning or retraining.

Although the scattering characteristics of the new scene differ from the training data, our model successfully suppressed the RFI and recovered terrain details with high fidelity. This result indicates that while the model has learned the scene-specific statistical distribution, the denoising process of DDPM has also learned a more fundamental generative representation of clean SAR signals. Consequently, even in unseen scenes, the model effectively utilizes spatial context to achieve reliable restoration. In comparison, other methods continued to perform inadequately. As shown in Table V, the proposed method significantly outperforms all compared methods in this cross-scene evaluation, robustly demonstrating its generalization capacity and practical utility.

3) Generalization to Different Satellite Platforms: Finally, to verify the cross-platform robustness of the proposed framework, we evaluated its performance on RADARSAT-1 satellite data. Since this dataset provides only single-polarization data, we used a channel duplication strategy, inputting the single-polarization time–frequency spectrogram into both network channels simultaneously. The model, pretrained on the Sentinel-1 simulation dataset, was directly used for testing without any fine-tuning. Qualitative suppression results are shown in Fig. 17. The results indicate that the proposed method outperforms the comparison methods across all metrics.

These results demonstrate that the proposed method achieves efficient RFI suppression and data restoration, even on data from a different satellite platform. This proves that the diffusion model does not overfit to specific training data parameters but instead captures the universal physical features of SAR echoes. Consequently, the model uses its strong spatial contextual inference ability to generalize effectively across different satellite platforms.

G. Performance on Real RFI-Contaminated Data

To validate the effectiveness of the proposed method in real-world scenarios, we used Sentinel-1 data contaminated with real RFI for our experiments. The RFI artifacts in the real scene, as shown in Figs. 18(a) and 19(a) are more complex due to amplitude variations and time deviations caused by transmission loss.

The experimental results in Figs. 18(b)–(f) and 19(b)–(f) again strengthen the conclusions observed with the simulated data. Traditional methods such as FNF, TFN, and RPCA perform poorly: while the filtering-based methods remove some interference, their inherent problem of data loss leads to a decrease in image quality. The performance of RPCA is unsatisfactory, leaving a large amount of residual interference, which strongly indicates that its reliance on the low-rank assumption is fragile in the face of complex, real RFI. The deep models IMN and PISNet also struggle to adapt to the complexity of real scenes. They not only leave significant residual RFI but also damage the useful signal, reducing image quality.

In contrast, our proposed framework demonstrates optimal robustness and effectiveness. As shown in Figs. 18(g) and 19(g), the method successfully suppresses the complex

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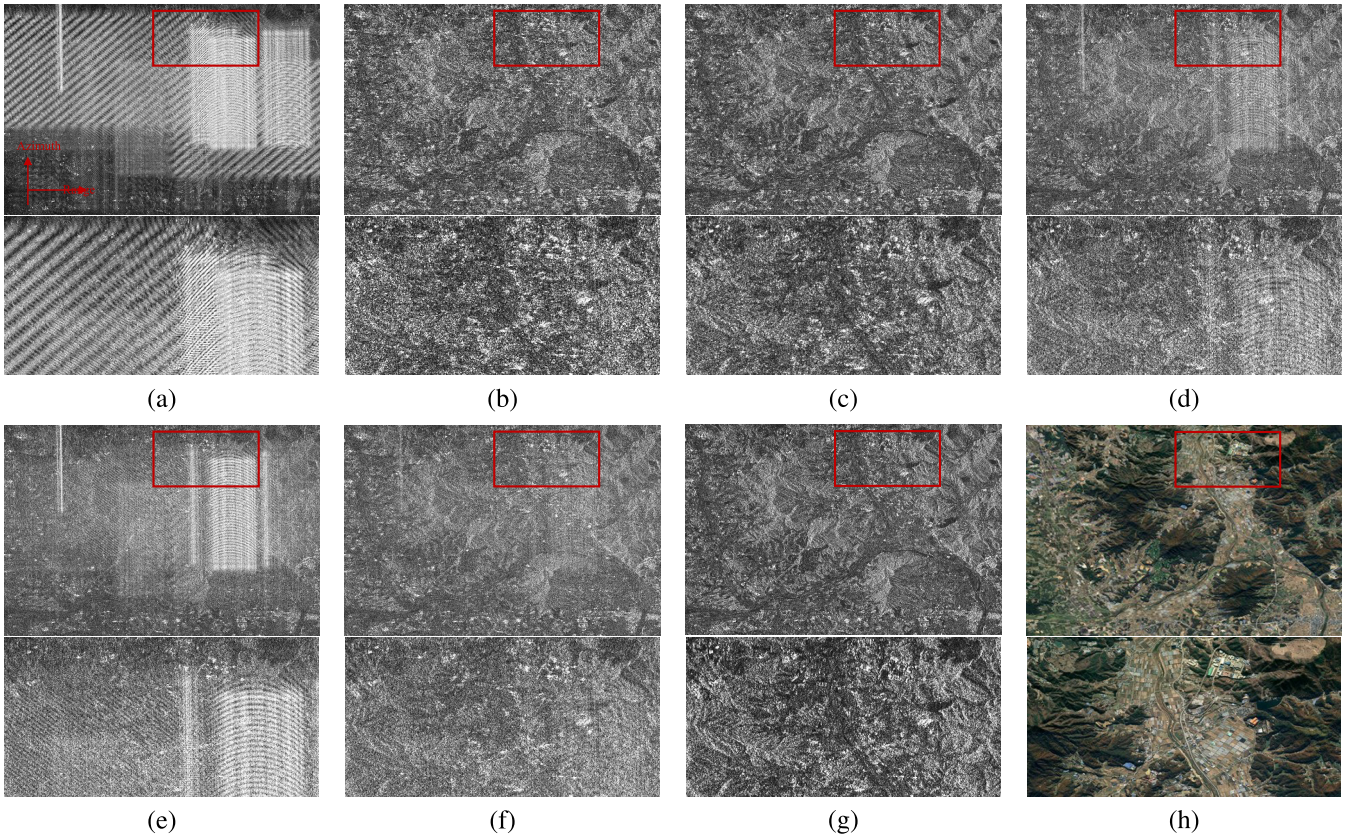

Fig. 18. Imaging results of selected region in real RFI contaminated data 1. The top row shows the selected regions, while the bottom row displays the corresponding ROIs from the areas indicated by the red boxes. (a) RFI-contaminated image. (b) FNF. (c) TFN. (d) RPCA. (e) IMN. (f) PISNet. (g) Proposed method. (h) Reference image from Google Earth.

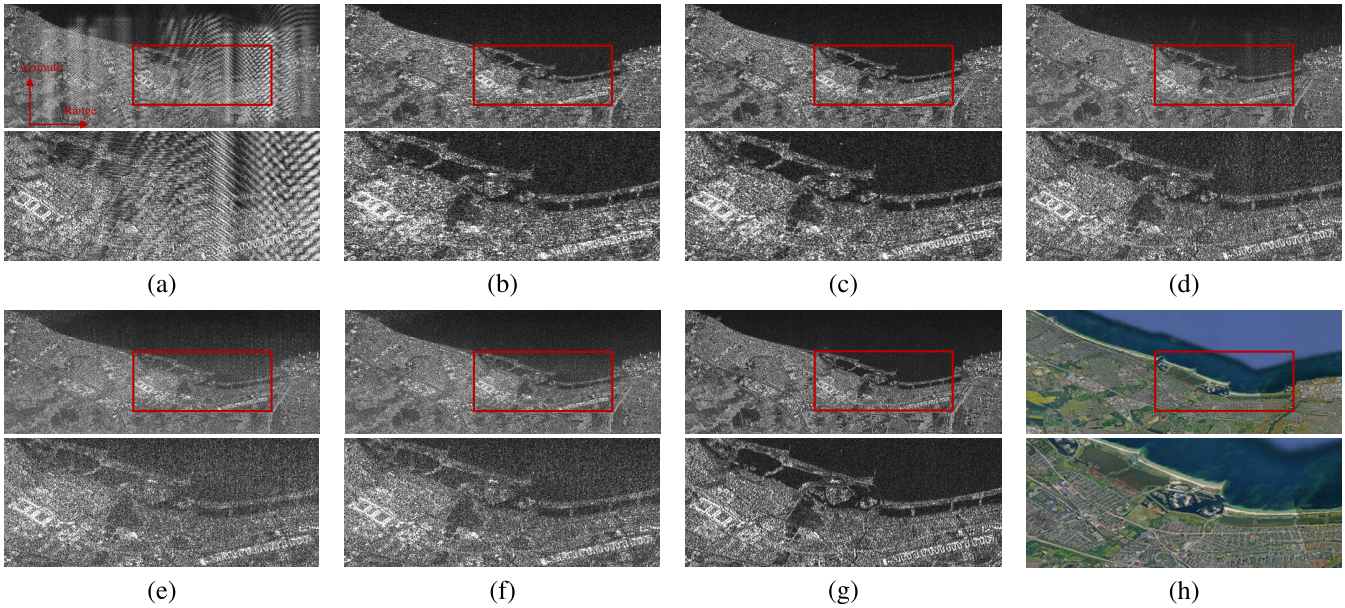

Fig. 19. Imaging results of selected region in real RFI contaminated data 2. The top row shows the selected regions, while the bottom row displays the corresponding ROIs from the areas indicated by the red boxes. (a) RFI-contaminated image. (b) FNF. (c) TFN. (d) RPCA. (e) IMN. (f) PISNet. (g) Proposed method. (h) Reference image from Google Earth.

real-world RFI while restoring the underlying scene with high fidelity. The final generated image is visually the clearest and most closely resembles the optical reference image provided by Google Earth. The no-reference image quality assessment metrics in Table VI further support this conclusion. Our method achieves the optimal on all metrics, fully demonstrating its superior performance in practical application scenarios.

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TABLE VI

QUANTITATIVE EVALUATION OF RFI SUPPRESSION PERFORMANCE ON A REAL SAR IMAGE. ↑ MEANS HIGHER VALUES ARE BETTER. ↓ MEANS LOWER VALUES ARE BETTER. Δ REPRESENTS THE AMOUNT OF CHANGE OF THE PROPOSED METHOD COMPARED TO THE SUBOPTIMAL

		With RFI	FNF	TFN	RPCA	IMN [35]	PISNet [37]	Proposed	Δ
Real Data 1	AG↑	50.70964	57.41338	60.52056	61.42320	60.41239	62.19154	**77.24074**	**+15.04920**
	GE↓	7.78141	7.69471	7.51483	7.54215	7.59125	7.47859	**7.41518**	**-0.06341**
	MSD↑	0.07785	0.06641	0.06459	0.06735	0.06573	0.06115	**0.08119**	**+0.01384**
Real Data 2	AG↑	45.70052	49.34951	51.25859	49.00332	45.51026	49.48142	**54.96723**	**+3.70864**
	GE↓	7.68386	7.54547	7.49418	7.51889	7.55633	7.54564	**7.43574**	**-0.05844**
	MSD↑	0.06977	0.07686	0.07218	0.07525	0.07137	0.07178	**0.07899**	**+0.00213**

V. CONCLUSION

In this article, we introduced a novel generative framework that reframes SAR RFI suppression as a conditional generation task. This approach represents a promising new direction for robust RFI mitigation and data-driven restoration. Our framework's core strength lies in the use of a conditional diffusion model to learn the underlying distribution of clean SAR signals. By sampling from this learned distribution, the model can perform a reliable and coherent restoration of contaminated data regions. A key innovation of our work is that this framework enables the model to handle complex and previously unseen RFI types, as the restoration is guided by the characteristics of clean signals, not by specific RFI patterns. To further enhance this generalization capability, we proposed a training strategy that utilizes a diverse library of physically simulated RFI masks. Experimental results on both simulated and real Sentinel-1 data validated the superiority of our method. Notably, our model demonstrated robust generalization, not only by effectively suppressing complex and unknown RFI but also by achieving excellent performance on data from geographical regions entirely unseen during training. However, it should be noted that the proposed method has limitations regarding computational efficiency. The iterative sampling process of the diffusion model results in relatively high computational costs compared to one-step deep learning models. Future work will focus on exploring accelerated sampling techniques to broaden its applicability in real-time processing scenarios.

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